

Manas Suple

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WORK EXPERIENCE

Mira-Bhayandar Municipal Corporation

Bhayandar

Intern

Jun 2023 - Jul 2023

- Maintained and organized operational data using spreadsheet-based record management systems.
- Assisted in digital workflow management and documentation processes across multiple departments.
- Supported website maintenance activities, including basic issue identification and content updates.
- Managed organizational communication channels, including email and social media platforms.
- Developed visual content and promotional materials using Canva and Adobe Premiere Pro.

PROJECT EXPERIENCE

Academic Project

AI-Based Conflict-Free Timetable Generation System

Jan 2026 - Present

Tech Stack: Python, FastAPI, PostgreSQL, SQLAlchemy, OR-Tools (CSP), Next.js, TypeScript

- Building a full-stack timetable management system that automatically generates conflict-free schedules across multiple departments and academic years.
- Implemented a Constraint Satisfaction Problem (CSP) based scheduling engine using Google OR-Tools to handle faculty availability, classroom allocation, lab continuity, and workload constraints.
- Built an independent validation module to detect faculty, room, and timetable conflicts before final approval.
- Designed REST APIs, database schemas, and file upload workflows for managing faculty, subject, classroom, and timetable data.
- Developed an admin dashboard for timetable generation, manual editing, validation, and PDF export with role-based access control.

Personal Project

RAG Backend + Streamlit App

Feb 2025 - Apr 2025

- Built a RAG-based PDF chatbot using LangChain, Google Gemini API, and Streamlit, enabling users to query documents in natural language with responsive, context-aware answers via ChromaDB.
- Engineered and deployed a modular backend for document parsing, chunking, embedding, and retrieval, with public access via Ngrok and future plans for multi-PDF support, multi-language queries, and cloud deployment.

AI-Powered Travel Assistant

Apr 2025 - Apr 2025

- Built an AI-powered travel planning agent using LangChain, Google Gemini, and WeatherAPI, delivering dynamic trip reports with weather, attractions, budgets, and flight details.
- Integrated real-time and simulated data sources to generate structured, user-friendly travel recommendations via a Python CLI, with planned upgrades including live flight APIs, hotel data integration, and exportable travel plans through an interactive Streamlit UI.

LinkedIn Post Generator

May 2025 - Jun 2025

- Developed a Streamlit-based LinkedIn post generator leveraging Google Gemini 2.0 and LangChain for context aware, professional content creation.
- Engineered an interactive web app to accept natural language inputs and generate fully formatted posts with hashtags and a consistent tone.

EDUCATION

K.J. Somaiya Institute of Technology, Mumbai

Mumbai

B.E. Electronics & Telecommunication (Ongoing)

Shri Bhagubhai Mafatlal Polytechnic, Mumbai

Mumbai

Diploma in Computer Engineering

Graduation Date: Mar 2024

CERTIFICATION

- Deloitte Data Analytics Virtual Experience Program — Forage (2025)
- Accenture Virtual Experience Program — Forage (2025)
- Capabl's Gen Ai Shaksham — KJ Somaiya (2025)
- Career Essentials in Generative AI — Microsoft & LinkedIn (2024)
- Cybersecurity Basics — KJ Somaiya (2024)
- Computer Networking & Coding Essentials — Scaler Topics (2024)

SKILLS & INTERESTS

Skills: Python, FastAPI, PostgreSQL, SQLAlchemy, REST API Development, LangChain, Google Gemini API, Next.js, TypeScript, React.js, Git, GitHub

Interests: Backend Development, AI Applications, Database Systems